

Region Segmentation based on Fluorescent Imaging

Anonymous CVPR submission

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Abstract

001 *Hospital-associated saliva deposition can seed high-risk*
002 *pathogens across shared surfaces. This work studies un-*
003 *supervised segmentation of saliva deposits illuminated un-*
004 *der UV-C, where only ten color images are available and*
005 *ground truth masks only cover four centralized patches per*
006 *frame. We explore a hierarchy of methods: a MATLAB-*
007 *based luminance/thresholding pipeline (Method A), an ex-*
008 *ploratory k-means segmentation (Method C), and our prob-*
009 *abilistic Gaussian mixture model (Method D) that reuses*
010 *k-means centers, runs ten EM initializations per image, and*
011 *selects the most likely fit by pseudo-log-likelihood. The re-*
012 *sulting GMM probability maps are binarized with morpho-*
013 *logical cleanup and evaluated on accuracy, precision, re-*
014 *call, and Dice; the $K=4$ model achieves the highest Dice*
015 *while maintaining recall above 0.96. Our comparison high-*
016 *lights how converting hard clusters into likelihood-ranked*
017 *mixtures improves recall while still recovering reasonable*
018 *precision, providing a practical unsupervised pipeline for*
019 *low-data airborne contamination imagery.*

020 1. Introduction

021 Hospital environments are vulnerable to pathogen transmis-
022 sion when saliva, which spreads easily between patients and
023 staff, settles on shared surfaces. Under UV-C illumination
024 these deposits glow brightly, but the dataset for segmenta-
025 tion is limited: ten color images spanning aluminum, steel,
026 white wood, PVC, and wood, with only four ground-truth
027 saliva masks per image. Thus the problem is best addressed
028 with unsupervised segmentation that can tolerate uneven
029 illumination, heterogeneous substrates, and tight recall re-
030 quirements rather than rely on large labeled datasets.

031 1.1. Problem definition

032 We formulate saliva segmentation as recovering centralized
033 bright regions that correspond to contamination while ig-
034 noring darker materials. The small size of the dataset and
035 the limited ground truth make supervised training unfeasi-
036 ble, so the methods must rely on image statistics and sim-

ple heuristics to propose saliva masks that can be validated
against the provided masks.

1.2. Contributions

In this work we deliver:

- A structured comparison of the existing MAT-
LAB/thresholding pipeline (Method A), the exploratory
k-means clustering pipeline (Method C), and the prob-
abilistic Gaussian mixture model pipeline (Method
D).
- A GMM workflow that seeds EM with k-means centers,
runs ten initializations per image, selects the best solu-
tion via pseudo-log-likelihood, and thresholds the result-
ing probability map before morphological cleanup.
- Empirical evidence that the $K = 4$ GMM increases
Dice while keeping recall above 0.96, illustrating how
likelihood-aware model selection improves the unsuper-
vised segmentation of saliva under UV-C.

2. Dataset and Methods

2.1. Dataset description

The dataset contains ten UV-C images collected over five
substrates (aluminum, steel, white wood, PVC, and wood)
along with binary ground-truth masks that highlight four
roughly centralized saliva patches per frame. The first eight
images serve as the tuning set while the remaining two are
held out for testing. Because the masks do not cover the
entire surface area and the total number of frames is small,
every method must operate in an unsupervised fashion and
rely on strong assumptions about color, intensity, and spa-
tial locality.

2.2. Method A: MATLAB thresholding pipeline

Method A is a lightweight computer-vision pipeline built in
MATLAB. Each RGB image is first converted to luminance
using ITU-R BT.601 weights; then Contrast Limited Adap-
tive Histogram Equalization (CLAHE) compensates for un-
even lighting. We apply Otsu's threshold with a learned bias
factor that shifts the threshold to include more saliva when
the raw threshold underestimates the foreground. Plausi-
bility filters remove regions whose area is either too small

(noise) or too large (non-saliva), and we reward regions that lie near the geometric center of the annotated patches. A final morphological cleanup (opening followed by closing) removes tiny speckles and fills holes created by the hard binarization.

2.3. Method B: placeholder

Method B was reserved for future work and is not implemented in this draft. The focus remains on documenting Methods A and C and on expanding the probabilistic mixture model described in Method D.

2.4. Method C: exploratory k-means

Method C runs k-means on the RGB pixels with $K = 3, 4, 5$ components. Each run is repeated multiple times (typically ten starts) and the best run is selected via the compactness metric (within-cluster sum of squares). We compare the luminance of each center ($0.114B + 0.587G + 0.299R$) and label the brightest one as saliva. Every pixel assigned to that center becomes foreground, and morphological opening/closing removes small artifacts while filling interior gaps. No preprocessing beyond the raw RGB conversion is needed because the UV-C illumination already maximizes contrast between saliva and background.

2.5. Method D: probabilistic Gaussian mixture model

Method D treats each pixel (R, G, B) as a sample from a Gaussian mixture. We fix $K \in \{3, 4\}$ to match the promising cluster sizes discovered by Method C and fit a full-covariance Σ_k GMM using `sklearn.mixture.GaussianMixture`. Because the EM algorithm only converges to a local optimum, the pipeline runs $n_{\text{runs}} = 10$ independent fits per image. The first few runs are seeded with k-means centers collected from Method C, while later runs use random initializations to cover diverse basins of attraction. Each run reports a pseudo-log-likelihood $\sum_i \log p(\mathbf{x}_i | \Theta)$ evaluated on the training data, and we keep the model with the highest likelihood as the most probable solution.

The selected model produces posterior probabilities $P(z_i = k | \mathbf{x}_i, \Theta^*)$. We identify the saliva component as the one with the highest luminance and extract $P_{\text{ig}}(\mathbf{x}_i)$. Thresholding this probability map at 0.5 yields a raw binary mask, which is then cleaned via morphological opening and closing to eliminate small false positives and fill holes. These masks are evaluated against the ground truth with accuracy, precision, recall, and Dice so that the results section can compare the methods on common metrics.

3. Results and discussion

Table 1 summarizes per-method averages for accuracy, precision, recall, Dice, and the pseudo-log-likelihood (PLL)

Method	K	Accuracy	Precision	Recall	Dice	PLL (10^6)
Method C (k-means, no preprocessing)	3	0.927	0.444	0.925	0.544	–
Method D (GMM)	3	0.930	0.282	0.992	0.432	-2.92
Method D (GMM)	4	0.959	0.419	0.964	0.564	-2.77

Table 1. Cross-method comparison of segmentation metrics. The GMM keeps recall high and the $K = 4$ solution improves Dice while increasing PLL relative to $K = 3$.

that Method D uses to select the best EM run. Method C maintains a high precision at $K = 3$, but the GMM keeps recall above 0.96 even when precision drops; switching to $K = 4$ raises Dice and improves the PLL, signaling a better probabilistic fit.

The probabilistic GMM absorbs bright saliva pixels that the hard clusters sometimes treat as background, yielding near-perfect recall at $K = 3$. The addition of a fourth component, selected by the highest PLL over ten EM runs (some seeded by k-means centers, some random), regains precision and raises Dice while still delivering high recall. This likelihood-guided selection, reinforced by the probability maps and morphological cleanup, ensures that the method reports not just predictions but also the fit quality described in ‘Leroy/results.txt’.

We will append runtime and memory profiling numbers once they are collected on the target hardware so the paper reports both segmentation performance and computational cost. For now, the emphasis is on demonstrating that Method D produces reliable masks under UV-C illumination despite the low-data regime.

4. Conclusion

This report documents an unsupervised saliva segmentation workflow: Method A’s MATLAB thresholding, Method C’s k-means clustering, and Method D’s likelihood-aware Gaussian mixture model. Method D keeps recall high and, at $K = 4$, raises Dice while using pseudo-log-likelihood to select the most probable model, demonstrating that probabilistic modeling and multi-start EM produce stable masks in this safety-critical, low-data setting.

References